

CS-INCLUDES
SCHEDULE PLANNER

REQUIREMENTS DOCUMENT

Prepared by
Project Team 2



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0 Revision History

Name	Date	Reason for Changes	Version
Ian Dang Allison Helling Dasanie Le Eugene Moshchyn Ricky Vincent Alan Vo	February 18, 2026	Initial Release	1.0
Ian Dang Allison Helling Dasanie Le Eugene Moshchyn Ricky Vincent Alan Vo	March 27, 2026	Sections 5, 6, 7, 8	2.0

1 Overview

The CS-INCLUDES SCHEDULE PLANNER is a web application that gives the CS-INCLUDES program at the University of Oklahoma a single place to manage all their events. Currently, mentors manually send separate emails and text messages to different mentee cohorts, which is very time-consuming and prone to mistakes. The new application fixes that issue by letting mentors create events, choose which mentees can see them, and automatically send reminders to mentees. Mentees get a personal calendar that only shows events meant for them, and signing up is as simple as clicking a button. If a mentee tries to sign up for overlapping events, the application detects the conflict, shows the events side by side, and asks the relevant mentor to approve before anything changes. Once a mentee signs up, the event is added to their Outlook or Google Calendar, and mentors are notified whenever a mentee cancels.

1.1 Team Members

List of team members:

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2 Business Requirements

2.1 Background

CS-INCLUDES is a program at OU that acts as an initiative designed to support and retain Native American and Indigenous students in the Computer Science program. They provide academic support, networking connections, professional development, and project opportunities. All while maintaining a family-oriented environment. To support their mission, CS-INCLUDES hosts a variety of weekly events, including resume workshops, study nights, and networking sessions.

2.2 Business Opportunity

CS-INCLUDES lacks a cohesive, dedicated method for organizing and scheduling the various events hosted. They need a dedicated application where mentors can plan, schedule, manage, and share events before they take place, so mentees can easily access and view both their options and currently scheduled events.

2.3 Business Objectives

The primary business objectives of this product are to improve operational efficiency, increase scholar engagement, and reduce manual administrative work for mentors. The implementation of this product should centralize 100% of event planning and communication into a single application and increase event attendance. The implementation of this product will benefit CS-INCLUDES by streamlining event management, making the experience seamless for event attendees, saving mentors time, and, in turn, boosting event attendance.

2.4 Success Metrics

Project success will be measured with the quantifiable measurements outlined in the business objectives, which include:

- The centralization of ALL event planning and communication into a single application.
- A notable numerical percentage increase in event attendance from past events.
- A numerical percentage decrease in the mentors' time spent manually planning.

The biggest factor in achieving success will be the centralization of event coordination. Meeting this key metric will help lead to the other success metrics being met.

2.5 Product Vision Statement

The CS-INCLUDES Schedule Planner will be a dedicated web application that streamlines the creation and organization of events run by the program. It will lessen the burden on mentors, allowing them to easily create events shared only with the relevant parties and accurately track who plans to attend. The planner will also assist mentees in keeping track of their schedules, providing a specialized view of their event calendar and helping them learn of new events. This product will replace the current manual email and text communication system, which is inefficient and tedious.

3 Scope and Limitations

3.1 Major Features

Mentors can create events with different visibility settings for mentees: invitation-only, open to everyone, or cohort-specific. Mentees can only view and join events relevant to them. When viewing events, information about event title, event description, date, time, and location will be displayed. Mentees also have access to the RSVP list to see the status of other attending mentees. If a mentee attempts to join an event that conflicts with their current schedule, an error message is displayed, and a notification is sent to their mentor. This scheduling conflict can be overridden by mentor approval. When a mentee joins an event, the event must be synced with their personal Google Calendar or Microsoft Outlook Calendar.

3.2 Project Scope

The CS-INCLUDES SCHEDULE PLANNER will centralize all program event operations into a single application for mentors and mentees. As mentioned in Section 3.1 - Major Features, mentors can create events with different visibility settings, and mentees can see and join events relevant to them. Ultimately, the CS-INCLUDES Schedule Planner will improve communication efficiency, communication clarity, and reduce the burden that mentors currently have from manually planning events.

3.3 Limitations and Exclusions

1. Incorporation of maps
2. Handling event recurrences
3. Wave-based reminders to the RSVP list
4. Integration with iOS
5. Two-way sync feature between Outlook Calendar and the application
6. Program admin interface

4 Context Description

4.1 User Classes and Characteristics

There will be three user roles: mentees, mentors, and a super admin. Mentees will have access to a calendar that shows the events they are eligible for. Mentees are lower-classmen with access to the product. Mentees will have the lowest security and privilege levels. Mentors will have access to create events and invite mentees and other mentors to the event. Mentors' security access is the same as mentees', with higher privileges, as they can create events and control the invite list. A super admin is a single person with a special ID who has the privilege to promote or demote user roles from mentee to mentor. The super admin will have the highest level of security access and privileges. The super admin has the highest level of education, as they are the head of the student organization.

4.2 Operating Environment

The application will exist as a website. The application must be compatible with any operating system that can access an internet browser. It must interact with the Microsoft Outlook API to sync with mentees' Microsoft Outlook calendars.

4.3 Design and Implementation Constraints

- **Role-Based Access Control:** Permissions are locked to each user and cannot be bypassed without higher-tier authorization
- **User Authentication:** Login is managed through the University of Oklahoma SSO Single Sign-On and Google's login service, so no custom login or encryption system needs to be built
- **Third-Party Integrations:** All reminders and calendar syncing route through Microsoft's Graph Application Programming Interface and Google Calendar's Application Programming Interface
- **Event Delivery Speed:** The system must support near-real-time delivery, with events appearing to eligible users within one minute of publishing
- **Data Storage:** Storage and server infrastructure should be simple and internally manageable, with no cloud database required
- **Technology Stack:** The client has no preference on programming language, framework, or database

4.4 Assumptions and Dependencies

Assumed factors that could affect the requirements stated in RD include interactions among the different roles. Should the super admin account also be a normal account? Can mentors also invite and view other mentor events? How much information should a mentee be able to see on the event page? The project depends on Microsoft Outlook calendars and Google calendars. Also, it is assumed that mentors and Mentees have a stable internet connection. Time conflicts are assumed to occur when an event overlaps with another event.

5 System Features

5.1 System Feature 1: Mentor Dashboard

5.1.1 Description

The mentor shall be able to create and edit events for mentees to view on their personalized calendars. Mentors can set event visibility so that mentees only see relevant events they are eligible for. (*Priority: High*)

5.1.2 Functional Requirements

- **MD-1:** Mentors shall create events with one out of three visibility settings (invitation-only, open-to-everyone, or cohort-specific).
 - Backward Traceability: This requirement is necessary to implement system feature 2, which displays only events that mentee is eligible for. This requirement was mentioned in Group 11 RFP Section 1.
 - Forward Traceability: This requirement can be verified by successfully creating an event with a certain visibility. Invitation-only events are visible for mentees only if they are invited by a mentor, open-to-everyone events are visible to all mentees, and cohort-specific events are visible if the mentee belongs to the cohort(s) that mentor specified.
- **MD-2:** Mentors shall be able to add info about event title, location, date, start time, end time, and description.
 - Backward Traceability: This requirement was specified by client in Client Meeting Notes which occurred on February 6, 2026.
 - Forward Traceability: This requirement can be verified by a mentee when they click on the event on their personal calendar and are able to see info about the event's title, location, date, start time, end time, and description.
- **MD-3:** Mentors shall be able to edit event title, location, date, start time, end time, and description.
 - Backward Traceability: Traceability: This requirement was specified by client in Group 11 RFP Section 2, Events Management Interface.
 - Forward Traceability: This requirement can be verified by the mentor successfully editing details of an event and having the changes reflected on mentees' personalized calendars.
- **MD-4:** Mentors shall be able to delete events.
 - Backward Traceability: This requirement was defined by Group 2 as a result of rendering this feature.
 - Forward Traceability: This requirement can be verified if the event gets deleted from the database

and all mentees' personal calendars.

- **MD-5:** Mentors shall have access to the RSVP list of events they are hosting.
 - Backward Traceability: This requirement was specified in Group 11 RFP Section 2, Mentor Interface.
 - Forward Traceability: This requirement can be verified if the application displays a list of event attendees when mentor clicks on any of their events from the mentor dashboard.
- **MD-6:** Mentors shall be able to invite mentees to their invitation-only events.
 - Backward Traceability: This requirement was specified in the Group 11 RFP, Glossary of terms section, under Mentor.
 - Forward Traceability: This requirement can be verified if a mentor can select mentees to invite from their dashboard. Personalized calendar should display the event as eligible only for mentees that have been invited.

5.1.1.3 Use Cases

UC-1: Create a Event	
Primary Actor	Mentor
Description	The mentor creates an event and sets visibility as either invitation-only, open-to-everyone, or cohort-specific. The mentor inputs info regarding event title, date, location, start time, end time, and description.
Trigger	Mentor indicates that they want to create an event from their dashboard.
Preconditions	PRE-1: User is authorized as mentor.
Postconditions	POST-1: New event is stored in a database. POST-2: New event displayed on calendars of mentees that satisfy visibility settings set forth by mentor.
Normal Flow (Open-to-everyone)	1. Mentor selects "Create Event" button on their mentor dashboard 2. Application displays empty event form 3. Mentor selects open-to-everyone visibility 4. Mentor inputs details regarding the event's title, location, date, start time, end time, and description 5. Mentor clicks "Publish Event" 6. Application saves event to database 7. Event is displayed as eligible on all mentees' personalized calendars
Alternative Flow 1.1 (Cohort-specific)	1. Mentor clicks "Create New Event" button on their mentor dashboard. 2. Application displays empty event form. 3. Mentor selects cohort-specific visibility. 4. Application provides additional input field for cohort (Freshman, sophomore, etc.) 5. Mentor inputs cohort, event title, location, date, start time, end time, and description. 6. Mentor clicks "Publish Event" 7. Application saves event to database 8. Event is displayed on personal calendars of mentees belonging to specified cohort
Alternative Flow 1.2 (Invitation-only)	1. Mentor clicks "Create New Event" button on their mentor dashboard. 2. Application displays empty event form. 3. Mentor selects invitation-only visibility 4. Application displays list of all mentees 5. Mentor selects mentees they would like to invite from list. 6. Mentor inputs event title, location, date, start time, end time, and description. 7. Mentor clicks "Publish Event" 8. Application saves event to database. 9. Event is displayed on personal calendars of invited mentees as eligible
Exceptions	If any one of the fields (visibility, title, location, date, start time, end time, and description) are missing, event creation will be blocked.
Priority	High

UC-2: Edit Event Details	
Primary Actor	Mentor
Description	A mentor can edit any information (event visibility, title, location, date, start time, end time, and description) of an event that they have created. After saving changes, the application should reflect these updates on mentees' personal calendars.
Trigger	Mentor indicates that they want to edit an event.
Preconditions	PRE-1: Mentor is the creator of this event.
Postconditions	POST-1: Event details are updated in the database. POST-2: Updated event details should be reflected on relevant mentee's personal calendars.
Normal Flow	1. Mentor selects an event from their dashboard. 2. Mentor inputs desired changes for the visibility, title, location, date, start time, end time, or description. 3. Mentor clicks "Save Changes". 4. Application updates event details in database. 5. Changes are reflected in relevant mentee's personal calendars.
Exceptions	If any field (visibility, title, location, date, start time, end time, description) is missing, event editing shall be blocked.
Priority	High

UC-3: Delete Event	
Primary Actor	Mentor
Description	Mentor can delete an event that they have created. The event should be deleted from the entire application.
Trigger	Mentor indicates that they want to delete an event.
Preconditions	PRE-1: Mentor is the creator of this event.
Postconditions	POST-1: Event is deleted from the database. POST-2: The event is deleted from the mentees' personalized calendars.
Normal Flow	1. Mentors select an event on their mentor dashboard. 2. Mentor clicks on "Delete Event". 3. Application deletes events from database. 4. Application removes event from all mentees' personalized calendars.
Priority	High

UC-4: View RSVP List	
Primary Actor	Mentor
Description	Mentors can view attendees on the RSVP list for an event that they have created.
Trigger	Mentor indicates that they want to view the RSVP list.
Preconditions	PRE-1: Mentor is the creator of this event.
Postconditions	POST-1: The application displays a list of attendees for an event.
Normal Flow	1. Mentors select an event on their mentor dashboard. 2. Application displays RSVP list.
Priority	High

5.1.4 User Stories

- ☐ **As a mentor**, I want to create events so that mentees can view them on their personalized calendars.
 - *Acceptance*: Mentor can access new event form after selecting "Create Event".
 - *Acceptance*: A mentor can create a new event that is displayed on personalized calendars of eligible mentees.
 - *Acceptance*: The application saves new event details to database.
- ☐ **As a mentor**, I want to control event visibility so that mentees see only the events that they are eligible to attend.
 - *Acceptance*: A mentor can set the visibility of an event as one of three settings (invitation-only, open to everyone, or cohort-specific).
 - *Acceptance*: A mentee's personal calendar only displays events that they are eligible for.
- ☐ **As a mentor**, I want to add event details so that mentees can make decision to RSVP or not.
 - *Acceptance*: New event form contains input fields for event's title, date, start time, end time, location, and description.
 - *Acceptance*: A event can not be created if it is missing event title, date, start time, end time, location, or description.
- ☐ **As a mentor**, I want to edit event details so that I can keep mentees updated on any changes.
 - *Acceptance*: A mentor can select an event they created and edit the event's title, date, start time, end time, location, or description.
 - *Acceptance*: A mentor can not save edits if any input field is missing.
 - *Acceptance*: After mentor saves changes, event details in database should be updated.
 - *Acceptance*: Editing changes should be reflected on mentee's personal calendars.
- ☐ **As a mentor**, I want to delete an event so that mentees know it is cancelled.
 - *Acceptance*: A mentor can select an event they have created and delete the event.
 - *Acceptance*: Event should be deleted from database.

- *Acceptance:* Event should be removed from all mentee's personal calendars.

□ **As a mentor**, I want to view the RSVP list so that I can monitor attendance and plan accordingly.

- *Acceptance:* A mentor can select an event they have created and access RSVP list.
- *Acceptance:* RSVP list should contain names of all mentees who have RSVP'd.
- *Acceptance:* RSVP list should not contain names of mentees who have un-RSVP'd.

5.2 System Feature 2: Mentee Personalized Calendar

5.2.1 Description

Mentees can access a personalized calendar that displays eligible and RSVP'd events. Mentees are able to view event details after clicking on an event. (*Priority: High*)

5.2.2 Functional Requirements

- **MPC-1:** Mentees shall have access to a personalized calendar that displays their eligible events and RSVP'd events.
- **MPC-2:** Eligible events shall be visually distinguishable from RSVP'd events on mentee's personalized calendar.
- **MPC-3:** Mentees shall be able to view an event's title, date, start time, end time, location, description, and RSVP list after clicking on event in personalized calendar.

5.2.3 Use Cases

UC-5: View Personalized Calendar	
Primary Actor	Mentee
Description	Mentee can view personalized calendar that displays eligible events and RSVP'd events. After clicking on event, mentee should be displayed with its information.
Trigger	Mentee navigates to their personalized calendar.
Preconditions	PRE-1: User is authenticated as mentee.
Postconditions	POST-1: Calendar filters and displays only events that mentee is eligible for. POST-2: Calendar displays mentee's RSVP'd events.
Normal Flow	1. Mentee navigates to their personalized calendar. 2. Personalized Calendar displays eligible events and RSVP'd events that are visually distinguishable. 3. Mentee clicks on the event. 4. Application displays a read-only view of event title, date, start time, end time, location, description, and RSVP list.
Priority	High

5.2.4 User Stories

- **As a mentee**, I want to view a personalized calendar, so that I can maintain a schedule of events I am eligible for and events I have RSVP'd.
 - *Acceptance:* A mentee has access to a personalized calendar.
 - *Acceptance:* Eligible events and RSVP'd events have different appearances when displayed on mentees' personalized calendars.

- **As a mentee**, I want to click on events so that I can see information that will help me decide whether I want to RSVP or not.
 - *Acceptance*: A mentee can view event title, date, start time, end time, location, description, and RSVP list after clicking on an event on their personalized calendar.

5.3 System Feature 3: Mentee RSVP/un-RSVP

5.3.1 Description

Mentees can RSVP and un-RSVP to events created by mentors and time conflicts will be handled. (*Priority: High*)

5.3.2 Functional Requirements

- **RSVP-1:** Mentees shall have the ability to RSVP to an event that they are eligible for.
- **RSVP-2:** Mentees shall have the ability to un-RSVP from an event.
- **RSVP-3:** Error message shall be displayed when a mentee attempts to RSVP an event whose time slot (start time to end time) overlaps with a RSVP'd events's time slot.
- **RSVP-4:** Mentors shall be able to approve a mentee's registration in a time-conflicting event.

5.3.3 Use Cases

UC-6: RSVP Event (with Time Conflict Handling)	
Primary Actor	Mentee
Secondary Actor	Mentor
Description	Mentees can RSVP to events that they are eligible for. A mentee may RSVP for an event that has time conflict with one of their currently RSVP'd events with mentor approval. The mentor will receive a notification on their dashboard that allows them to approve or reject the request.
Trigger	Mentee indicates that they want to RSVP to an event.
Preconditions	PRE-1: Mentee is eligible for this event (event is displayed on their personalized calendar).
Postconditions	POST-1: Mentee is added to event's RSVP list. POST-2: Event is marked on mentee's personalized calendar as RSVP'd.
Normal Flow (No Conflicts)	1. Mentee selects an eligible event on their personalized calendar. 2. Mentee clicks on "RSVP". 3. Mentee gets added to event RSVP list. 4. Application marks event as RSVP'd on mentee's personal calendar.
Alternative Flow 6.1 (Conflict Approved)	1. Mentee clicks on eligible event on their personalized calendar. 2. Application displays time conflict warning and registration is blocked. 3. Mentee clicks on "Request Mentor Approval" button. 4. Notification is sent to the mentor's dashboard. 5. The mentor approves the registration. 6. Mentee is added to the new event's RSVP list in the database. 7. Application marks new event as RSVP'd on mentee's personal calendar.
Alternative Flow 6.2 (Conflict Rejected)	1. Mentee clicks on eligible event on their personalized calendar. 2. Application displays time conflict warning and registration is blocked. 3. Mentee clicks on "Request Mentor Approval" button. 4. Notification is sent to mentor's dashboard. 5. Mentor rejects the registration. 6. Application sends rejection notification to mentee and no changes are made to the mentee's personalized calendar.
Priority	High

UC-7: Un-RSVP Event	
Primary Actor	Mentee
Description	Mentees can un-RSVP to any currently RSVP'd events. The mentees' personalized calendar must reflect these changes.
Trigger	Mentee indicates that they want to un-RSVP from an event.
Preconditions	PRE-1: Mentee has already RSVP'd to an event.
Postconditions	POST-1: Event is marked as eligible on the mentee's personalized calendar. POST-2: Mentee is removed from event's RSVP list.
Normal Flow	1. Mentee selects event that they have already RSVP'd. 2. Mentee selects "Un-RSVP" button. 3. Event is marked as eligible on mentee's personalized calendar. 4. Mentee is removed from event's RSVP list in database.
Priority	High

5.3.4 User Stories

- ☐ **As a mentee**, I want to RSVP events so that I can secure my spot.
 - *Acceptance:* A mentee can RSVP to an event that is marked as eligible on their personalized calendar.
 - *Acceptance:* A mentee is saved to event's RSVP list in database.
 - *Acceptance:* After mentee RSVPs, event is visually distinguishable from other eligible events on their personal calendar.
- ☐ **As a mentee**, I want to un-RSVP from events, so that my mentor knows I no longer want to attend.
 - *Acceptance:* A mentee can un-RSVP from an event[cite: 399].
 - *Acceptance:* A mentee is removed from event's RSVP list in database.
 - *Acceptance:* After mentee un-RSVPs, their personal calendar should mark the event as eligible.
- ☐ **As a mentee**, I want to request mentor approval for an overlapping event so that I can possibly attend both.
 - *Acceptance:* A mentee can request permission to override time conflict after clicking on event details from their personalized calendar.
 - *Acceptance:* Application marks time-conflicting event as RSVP'd if mentor approves their RSVP request.
- ☐ **As a mentor**, I want to receive time conflict notifications, so that I can approve or reject a mentee's RSVP request for an event with a time conflict.
 - *Acceptance:* A mentor can receive a time conflict notification when a mentee attempts to RSVP for an event that overlaps with their RSVP'd events.
 - *Acceptance:* A mentor can approve or reject a mentee's RSVP request for a time conflicting event.

5.4 System Feature 4: Super Admin Interface

5.4.1 Description

Super Admin Interface allows the super admin to assign roles to users as mentees or mentors. The super admin can also promote or demote users. (*Priority: High*)

5.4.2 Functional Requirements

- **SAI-1:** Super Admin shall be able to assign mentor/mentee roles to users.
- **SAI-2:** Super Admin shall be able to edit user roles.

5.4.3 Use Cases

UC-8: Assign User Role	
Primary Actor	Super Admin
Description	Super Admin is able to promote, demote, and assign roles (mentee or mentor) to users.
Trigger	Super Admin clicks on "Edit Role" on their dashboard.
Preconditions	PRE-1: User is authorized as Super Admin. PRE-2: Targeted user has logged in via OU SSO and database has saved "unsigned" to their role field.
Postconditions	POST-1: User's new role is assigned to the role field in database.
Normal Flow	1. Super Admin navigates to their Super Admin interface. 2. Super Admin selects targeted user. 3. Super Admin selects "Edit Role". 4. Super Admin selects desired role (Mentor or Mentee) from drop down menu. 5. Super Admin selects "Save Changes".
Priority	High

5.4.4 User Stories

- **As a super admin**, I want to assign roles to users so that users have access to their correct interfaces.
 - *Acceptance:* A Super Admin can access their Super Admin dashboard.
 - *Acceptance:* A Super Admin can assign mentee or mentor role to user.
 - *Acceptance:* A Super Admin can promote or demote users between mentors and mentees.

5.5 System Feature 5: External Calendar Syncing

5.5.1 Description

After a mentee RSVP's for an event, the event should be automatically synched with Google Calendar or Microsoft Outlook Calendar. (*Priority: Medium*)

5.5.2 Functional Requirements

- **ECS-1:** RSVP'd events shall be automatically integrated into Google Calendar and Microsoft Outlook Calendar.
- **ECS-2:** Google and Microsoft Outlook Calendars shall contain same event title, date, start time, end time, location, and description as application.

5.5.3 Use Cases

UC-9: Sync Event to External Calendar	
Primary Actor	Mentee
Secondary Actor	Microsoft Graph API and Google Calendar API
Description	After a mentee RSVPs to an event, the application will automatically push event details to their Google Calendar or Microsoft Outlook Calendars.
Trigger	Mentee RSVPs an event.
Preconditions	PRE-1: Mentee is RSVP'd to an event. PRE-2: Mentee has linked their Google or Microsoft Outlook calendar(s).
Postconditions	POST-1: Event is successfully integrated into external calendar.
Normal Flow	1. Mentee RSVP's to event. 2. Application sends event details to external calendar APIs. 3. Event is created on external calendar.
Priority	Medium

5.5.4 User Stories

- **As a mentee**, I want my RSVP'd events to automatically sync to my Google/Microsoft Calendars so that I don't have to manually add the event.
 - *Acceptance:* After a mentee RSVPs, the event is displayed on their external calendar.
 - *Acceptance:* Events on application and external calendar share the same event title, date, start time, end time, location, and description.

6 External Interface Requirements

6.1 User interfaces

6.1.1 User Interface Components

- **Mentor Dashboard:** Facilitates event creation and management.
- **Mentee Personal Calendar:** Provides a specialized view of eligible events and currently RSVP'd events.
- **Super Admin Interface:** Allows a designated user to promote or demote user roles.
- **Event details/RSVP Page:** Displays event specifics and information, allowing for registration or viewing the RSVP list.

6.1.2 Interface Characteristics

- **Centralized navigation:** The application will have a central portal for all event planning and communication.
- **Interactive Calendar View:** The mentees will have access to an interactive calendar where clicking a date displays information about the event.
- **Standard buttons:** Standard buttons for event management and interaction for both mentees and mentors.

6.1.3 GUI Standards

- **Responsive Web Design:** The website should be compatible with modern operating systems and devices.
- **Error message display standards:**
 - **Time Conflict Displays:** If a mentee attempts to join an event that overlaps with an existing event scheduled, the system will display an error message.
 - **Mentor Notifications:** Notifications are automatically sent to mentors when a mentee cancels an event or when a scheduling conflict requires mentor approval.

6.2 Hardware interfaces

Since the scheduler is primarily web-based, it will have minimal specialized hardware requirements.

The hardware requirements for users are a reliable internet connection and access to a supported internet browser.

Server-side, the application requires hosting hardware that will support real-time event publishing, user

authentication requests, database operations, and calendar synchronization requests.

6.3 Software interfaces

- **OU SSO and Google Login/OAuth:** For institutional user authentication and Google account authentication.
- **Microsoft Graph API and Google Calendar API:** For calendar synchronization.
- **Data Storage Layer:** For shared data to remain consistent across the frontend, backend, database, and third party integrations.
- **Communication between frontend web interface and backend application server:**
 - Frontend sends requests (eg. for viewing events, creating events, RSVPing, managing user roles, etc.).
 - Backend processes the requests, applies business rules, stores/retrieves data from the database, and communicates with APIs for calendar synchronization or authentication.

6.4 Communication interfaces

- **Web communication:**
 - Client-server communication occurs over HTTPS for secure data transmission.
 - The system will support modern web browsers.
- **API communication:**
 - Microsoft Graph API to synchronize our platform's mentee's calendars with their microsoft outlook calendars.
 - We will have our own backend API endpoints to handle requests from the frontend.
- **Authentication communication:**
 - OU SSO.
 - Google Login/OAuth to allow for google account authentication.
- **Notification communication:**
 - Mentors are sent a notification from the system if scheduling conflict happens.
 - Mentees are sent a notification from the system if attempting to RSVP for time conflicting event.
- **Frontend-Backend communication:**

- Expose Backend API endpoints and configure CORS for the backend server to allow access only from trusted frontend origins.

7 Software Quality Attributes

7.1 Performance

- **PERF-1:** When a user logs in, application shall display their respective interfaces (Mentee Personalized Calendar or Mentor Dashboard) within 2 seconds.
- **PERF-2:** When an event is created, application shall display it on eligible mentees' personalized calendars within 2 seconds.
- **PERF-3:** When mentor edits an event, application shall update mentees' personalized calendars within 2 seconds.
- **PERF-4:** When mentor deletes an event, application shall remove it from mentees' personalized calendars within 2 seconds.
- **PERF-5:** When mentee RSVPs or un-RSVPs from an event, application shall update their personalized calendar within 2 seconds.
- **PERF-6:** When mentee requests approval to RSVP a time-conflicting event, a notification shall be sent to mentor's dashboard within 2 seconds.

7.2 Availability

- **AV-1:** The application shall be available to all users for at least 99.9% of the time.

7.3 Security

- **SEC-1:** There shall be no exposure of confidential information (PII).
- **SEC-2:** Mentors shall only have access to their mentor dashboard.
- **SEC-3:** Mentees shall only have access to their personalized calendar.
- **SEC-4:** Super Admin is only user who can promote and demote other users.
- **SEC-5:** User authentication shall be handled by OU SSO and Google Login/OAuth.

7.4 Usability

- **USAB-1:** RSVP'd and eligible events shall have different appearances on a mentee's personalized calendar for clear distinction.
- **USAB-2:** Mentor can perform event creation, deletion, editing, invitations, and time-conflict handling from their mentor dashboard with at most 5 button clicks (excluding input field clicks).
- **USAB-3:** Mentees can RSVP/un-RSVP on their personalized Calendar with at most 3 button clicks

(excluding input field clicks).

8 Analysis Models

Use Case Diagram

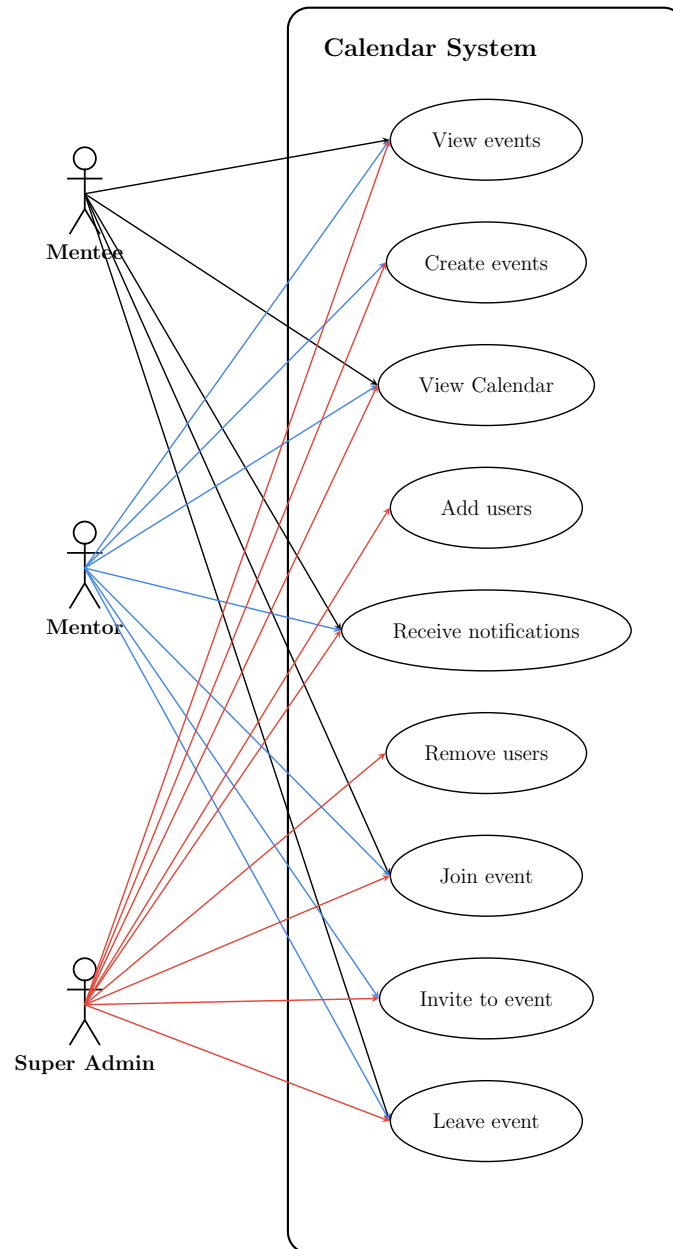


Figure 1: Use Case Diagram

Architecture Diagram

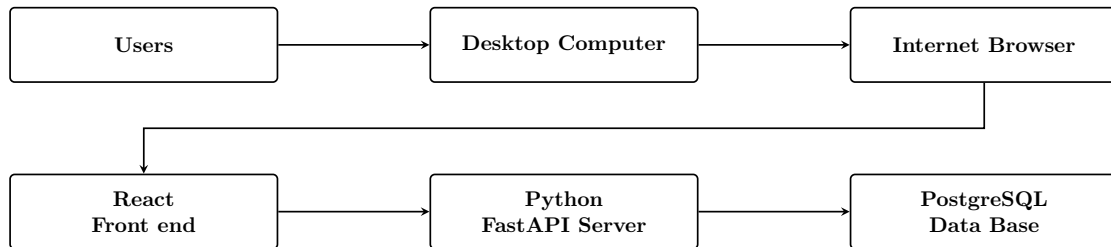


Figure 2: High-Level System Architecture Diagram

Context Level Data Flow Diagram

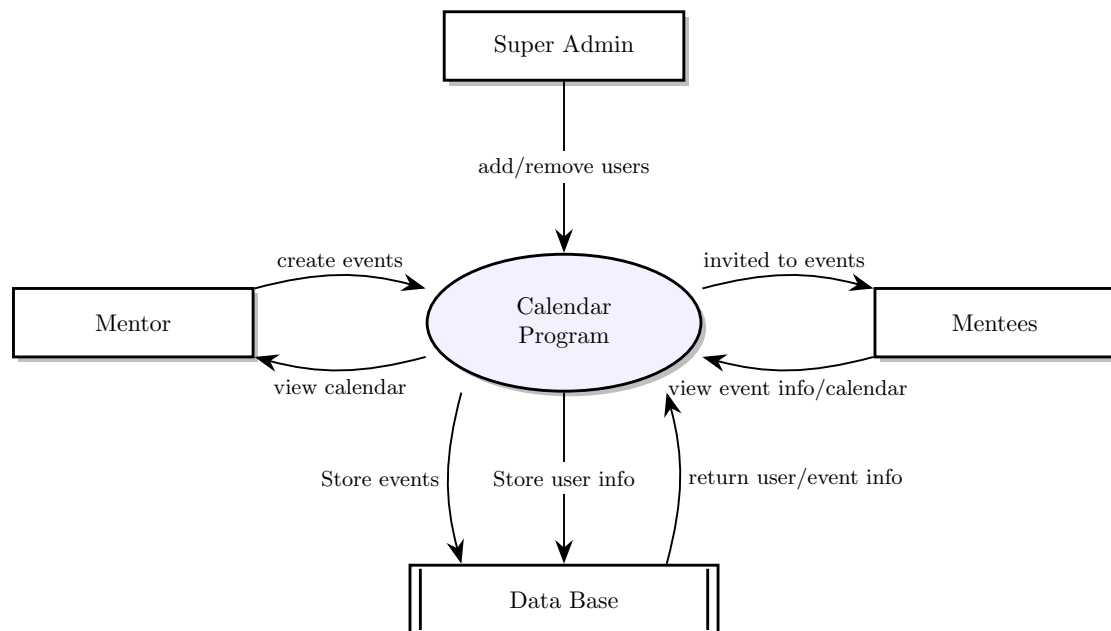


Figure 3: Context-Level Data Flow Diagram

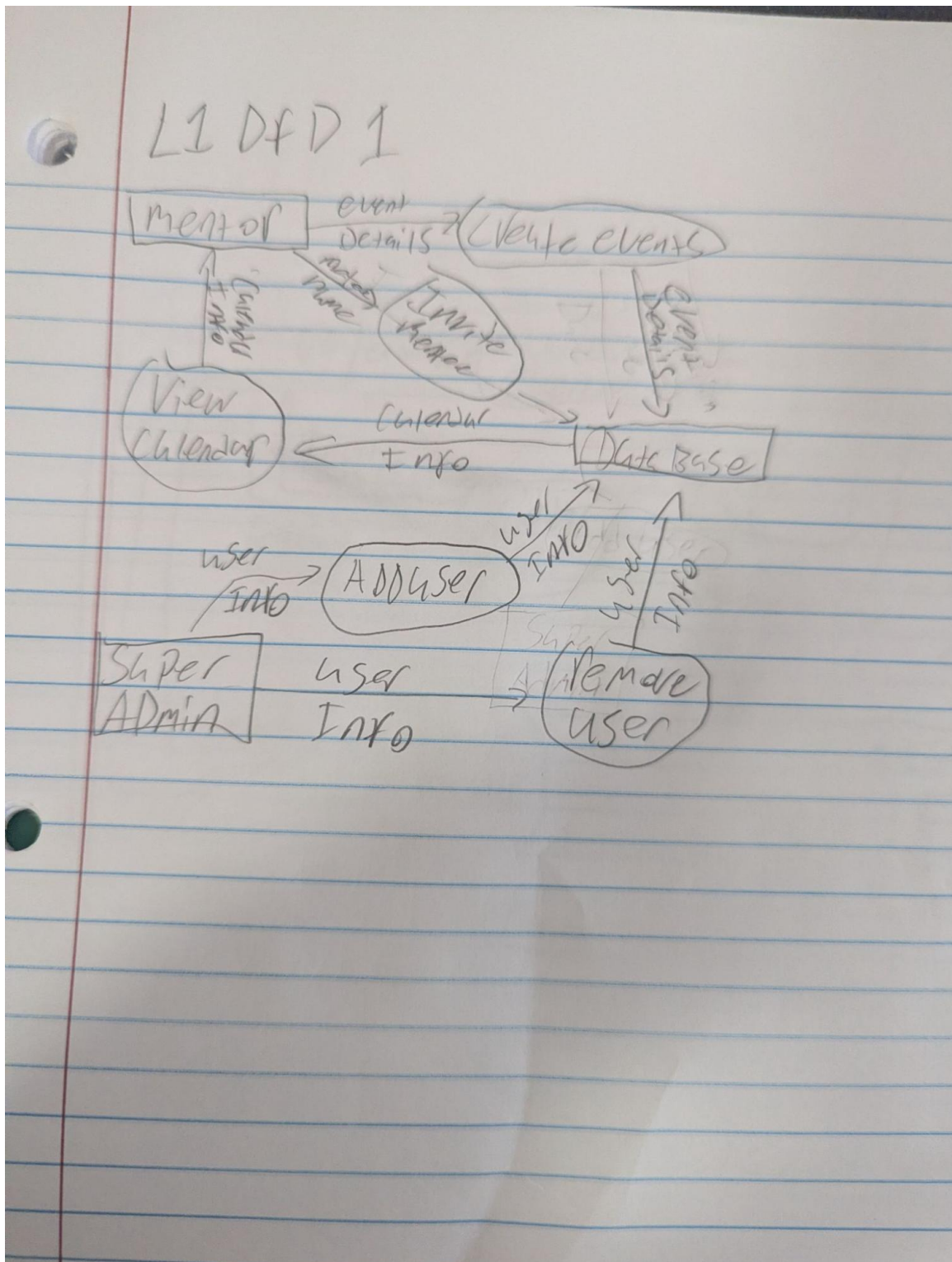


Figure 4: Level 1 Data Flow Diagram 1

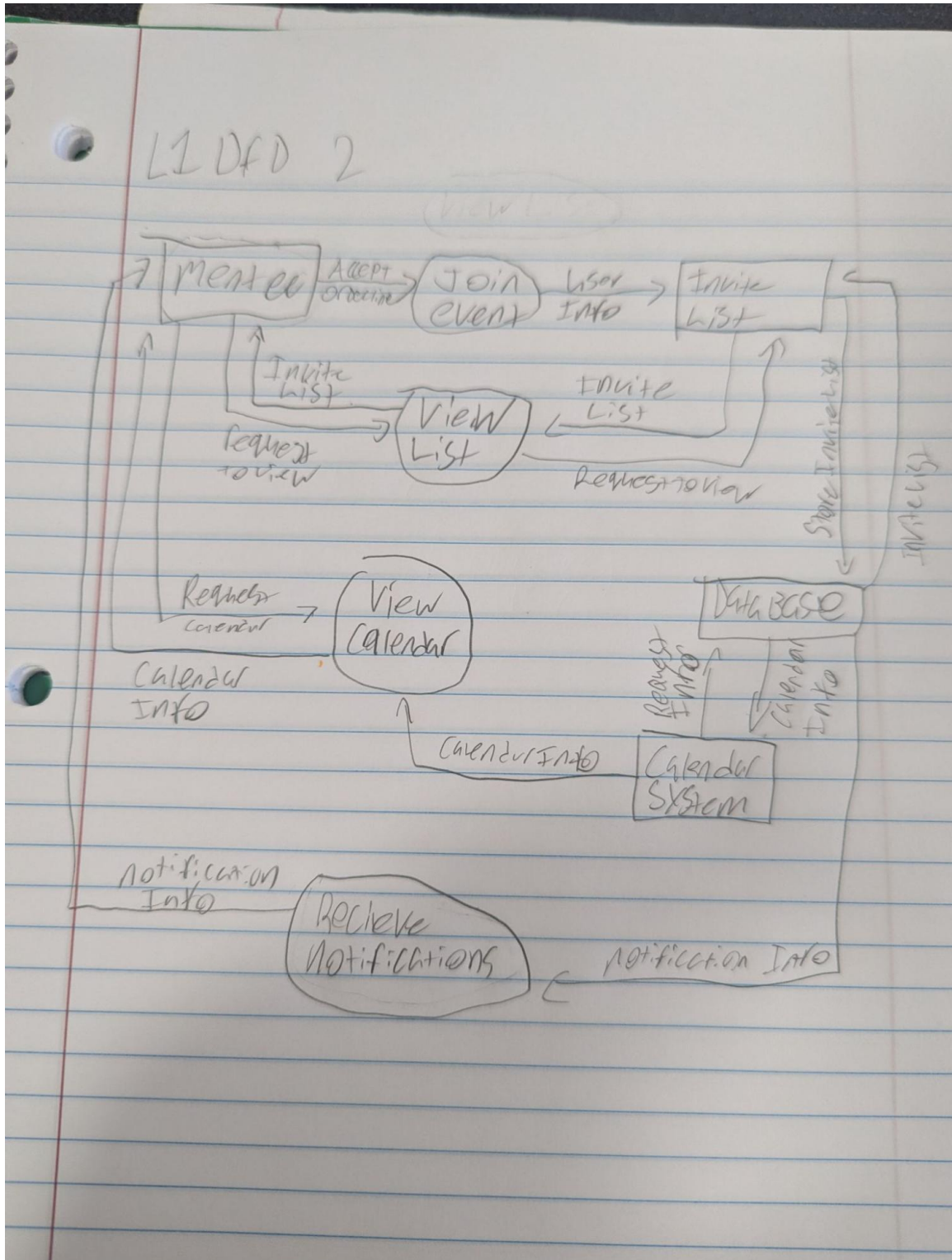


Figure 5: Level 1 Data Flow Diagram 2

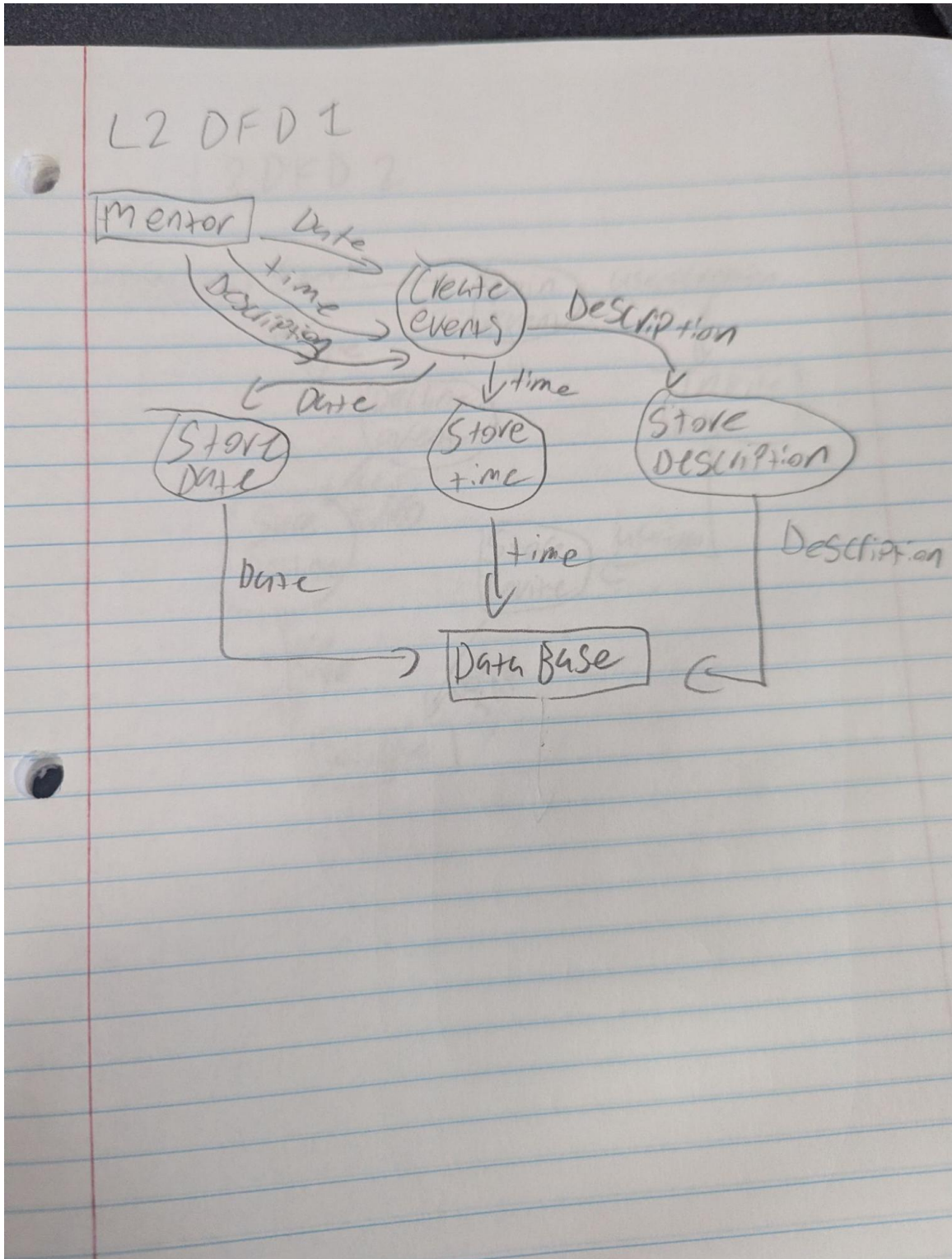


Figure 6: Level 2 Data Flow Diagram 1

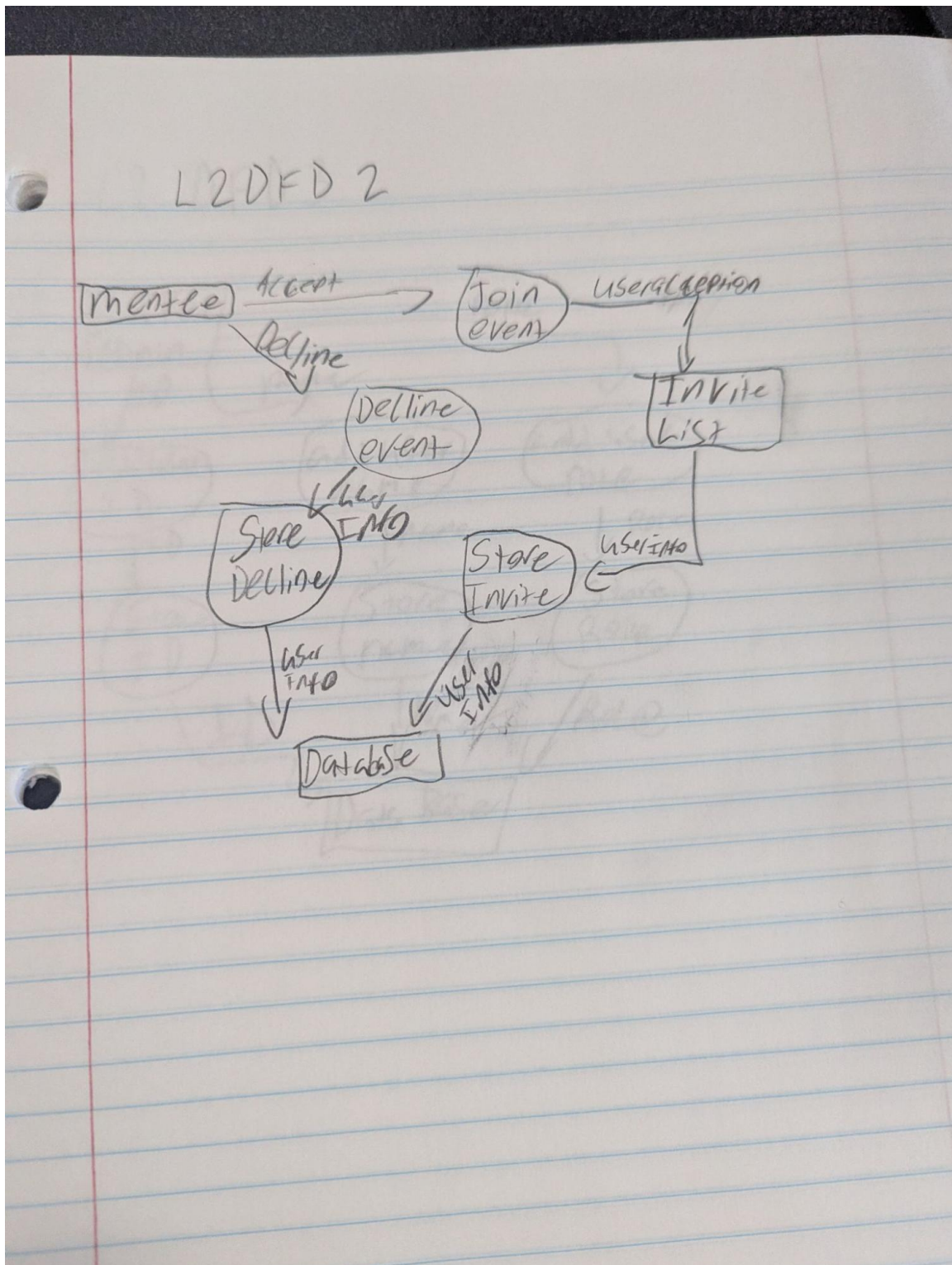


Figure 7: Level 2 Data Flow Diagram 2

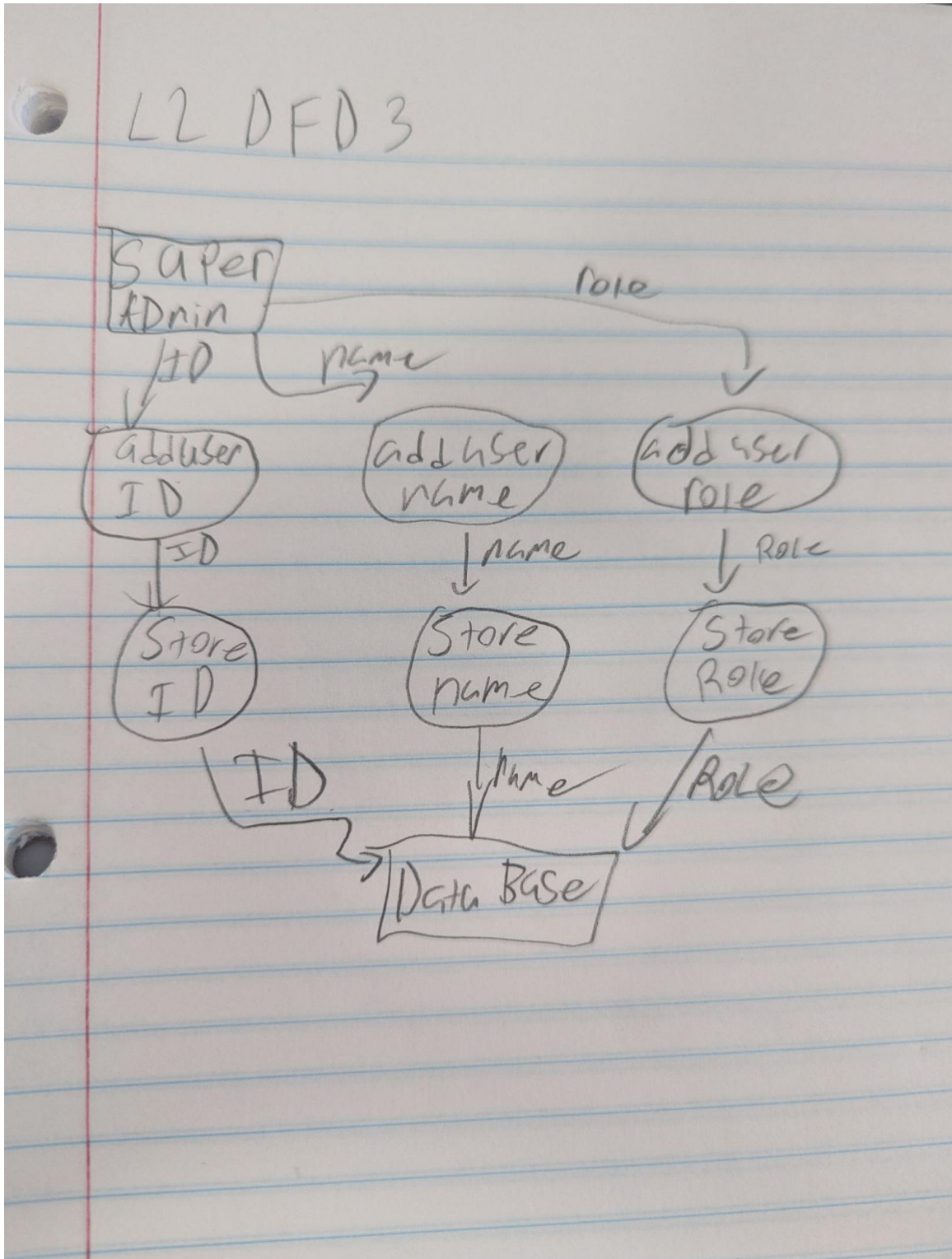


Figure 8: Level 2 Data Flow Diagram 3

A Glossary of Terms

1. **Super Admin** — The leader of the student organization who can promote and demote members
2. **Mentor** — Upperclassmen responsible for creating and organizing events
3. **Mentee** — Lower-classmen who will be invited to events
4. **Cohort** — A group of mentees in the same academic year, such as Freshman, Sophomore, Junior, Senior

B Reflection on the Use of AI

B.1 Tools and Models Used

1. ChatGPT by OpenAI
2. Gemini by Google

B.2 Prompt History

1. ChatGPT: “Provide a glossary of terms based on the provided document”
2. Gemini: “Explain peer to peer isolation and how to implement it for a mentor class in plain English”
3. ChatGPT: “What is role-based access control?”
4. ChatGPT: “What is Microsoft Graph API?”
5. ChatGPT: “What is backward and forward traceability?”

B.3 How AI Support the Work

AI tools were used selectively to accelerate specific tasks rather than to generate the document as a whole. ChatGPT was used to produce an initial list of terms for the glossary and to clarify technical concepts such as role-based access control, the Microsoft Graph API, and traceability practices. Gemini was used as a research tool to help explore permission structures. In all cases, AI output served as a reference rather than a final product.

B.4 Critical Evaluation of AI Effectiveness and Limitations

AI tools were effective for clarifying technical concepts and generating starting points for specific tasks. However, AI models have no knowledge of the CS-INCLUDES program, its user base, or its constraints, meaning all output required careful review and refinement by the team before incorporation.

B.5 Student Judgement and Understanding

The team used AI as a tool for efficiency. Every AI-generated output was reviewed before being incorporated. The team is aware that AI models can produce inaccurate information, so all outputs were treated with appropriate skepticism. The design decisions in this document reflect the team’s own analysis of the client’s needs.

References

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- [3] Group 11, University of Oklahoma. Cs-includes schedule planner request for proposal. Version 0.1, February 2026. Provided directly by the client to Group 11.
- [4] Group 2. Client meeting notes, design and requirements clarification, February 2026. Internal document, available upon request from Group 2.
- [5] Ze Shi Li. Cs-includes schedule planner requirements document template. Version 1.0, University of Oklahoma, February 2026. Provided via course materials.
- [6] Microsoft Corporation. *Microsoft Graph API Reference*. Continuously updated.
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